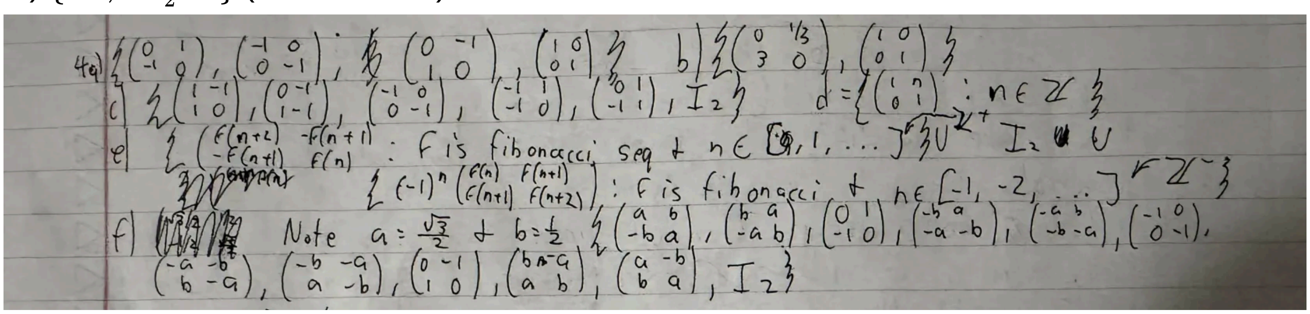
hw3

Ch 4

1. a) No, 1 generates all of $\mathbb{Z}_{60}$ and 1 is not prime
   b) No U(8) is isomorphic to the Klein-4 group which is not cyclic
   Additionally, U(8) = {1, 3, 5, 7}. 1 does not generate, 3^2 mod 8 = 1 so 3 does not generate. 5^2 mod 8 = 1 so 5 does not generate.y^2 mod 8 = 1 so 7 does not generate.
   Therefore U(8) is not cyclic
   c) No, proof by contradiction. Let $\mathbb{Q} =< a >$
   Then $\forall x \in \mathbb{Q}, x = na$ where $n \in \mathbb{Z}$. Since $a \in \mathbb{Q}, a = \frac{p}{q}$ where $p, q \in \mathbb{Z}$ and gcd(p,q)=1. Let x = P where P is a prime bigger than any prime factor of p. Then $P = \frac{np}{q}$, since n is an integer, this would imply that P has factors which it can't since we chose it to be a prime.
   Therefore contradiction
   d) No look at U(8) from part b. It is not cyclic but every subgroup is cyclic
   e) No. Assume group G has finite amount of subgroups. If one of the subgroups is infinite, then it is isomorphic to $\mathbb{Z}$ and therefore has infinite subgroups so if G has finite amount of subgroups, the size of the subgroups must be finite therefore G must be finite as well.
2. a) 12
   b) inf
   c) inf
   d) 4
   e) 240/gcd(72,240)=240/24=10
   f) gcd(312,471)=gcd(312,159)=gcd(159,153)=gcd(6,153)=gcd(6, 3)=3 so 471/3=157
3. a) all mult of 7 ie $\{7n : n \in \mathbb{Z}\}$
   b) {15, 6, 21, 3, 18, 9, 12, 0}
   c)
   <0>= {0}
   <1>= {0, ... , 11}
   <2>= {0, 2, 4, 6, 8, 10}
   <3>= {0, 3, 6, 9}
   <4>= {0, 4, 8}
   <6>= {0, 6}
   d)
   <0>= {0}
   <1>= {0, ... , 59}
   <2>= {0, 2, 4, 6, 8 ... 58}
   <3>= {0, 3, 6, 9 ... 57}
   <4>= {0, 4, 8 ... 56}

<5>= {0, 5 ... 55}
<6>= {0, 6 ... 54}
<10>= {0, 10 ... 50}
<12>= {0, 12 ... 48}
<15>= {0, 15 ... 45}
<20>= {0, 20, 40}
<30>= {0, 30}
e) Trivial subgroups, {0} and $\mathbb{Z}_{13}$
f)
<0>= {0}
<1>= {0, ... , 47}
<2>= {0, 2, 4, 6, 8 ... 46}
<3>= {0, 3, 6, 9 ... 45}
<4>= {0, 4, 8 ... 44}
<6>= {0, 6 ... 42}
<8>= {0, 6 ... 40}
<12>= {0, 12, 24, 36}
<15>= {0, 16, 32}
<24>= {0, 24}
g) {3, 9, 7, 1}
h) {5, 7, 17, 13, 11, 1}
i) $\{7^n : n \in \mathbb{Z}\}$
j) {i, -1, -i, 1}
k) $\{(2i)^n : n \in \mathbb{Z}\}$
l) $\{\frac{1+i}{\sqrt{2}}, i, \frac{1+i}{\sqrt{2}}i, -1, -\frac{1+i}{\sqrt{2}}, -i, -\frac{1+i}{\sqrt{2}}i, 1\}$
m) $\{\pm 1, \frac{\pm 1 \pm i\sqrt{3}}{2}\}$ (6 total elems)

4. 

4a) $\{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\}$ b) $\{\begin{pmatrix} 0 & 1/3 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\}$

c) $\{\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, I_2\}$ d $= \{\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z}\}$

e) $\{\begin{pmatrix} F(n+2) & -F(n+1) \\ -F(n+1) & F(n) \end{pmatrix}$: F is fibonacci seq & $n \in [0, 1, \dots] \leftarrow \mathbb{Z}^+\} \cup I_2 \cup$ $\{(-1)^n \begin{pmatrix} F(n) & F(n+1) \\ F(n+1) & F(n+2) \end{pmatrix}$: F is fibonacci & $n \in [-1, -2, \dots] \leftarrow \mathbb{Z}^-\}$

f) Note $a = \frac{\sqrt{3}}{2}$ & $b = \frac{1}{2}$ $\{\begin{pmatrix} a & b \\ -b & a \end{pmatrix}, \begin{pmatrix} b & a \\ -a & b \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} -b & a \\ -a & -b \end{pmatrix}, \begin{pmatrix} -a & b \\ -b & -a \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -a & -b \\ b & -a \end{pmatrix}, \begin{pmatrix} -b & -a \\ a & -b \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} b & -a \\ a & b \end{pmatrix}, \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, I_2\}$

5. |<1>| = |<17>| = |<5>| = |<13>| = |<7>| = |<11>| = 18
|<2>| = |<16>| = |<4>| = |<14>| = |<8>| = |<10>| = 9
|<3>| = |<15>| = 6
|<6>| = |<12>| = 3
|<9>| = 2
|<0>| = 1

6. Elements r_{90}, r_{180}, r_{270} which are rotations, D_1, D_2 which are diagonal flips, e which is stay same, V and H which are vertical and horizontal flips
|<r_{90}>| = |<r_{270}>|= 4
|<r_{180}>| = |<D_1>| = |<D_2>| = |<V>|= |<H>| = 2
|<e>| = 1
7. Q_8 = {1, -1, i, -i, j, =j, k, -k}
<1>={1}
<-1>={-1, 1}
<i>={i, -1, -i, 1}
<j>={j, -1, -j, 1}
<k>={k, -1, -k, 1}
8. U(30) = {1,7,11,13,17,19,23,29}
<1> = {1}
<7> = {1, 7, 19, 13}
<11> = {1, 11}
<13> = {1, 13, 19, 7}
<17> = {1, 17, 19, 23}
<19> = {1, 19}
<23> = {1, 17, 19, 23}
<29> = {1, 29}
<11, 29> = {1, 11, 19, 29}

Ch 6

1. Because if g has order 5 and h has order 7, By lagrange's theorem the order of a subgroup must divide the order of the group so 5 and 7 divide G and the smallest such number with 5 and 7 as factors is 35 so |G| >= 35
2. 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60
3. False, let G= $\mathbb{Z}$ and H= {0}, the trivial subgroup. H has infinite index as we can take any element $x \in \mathbb{Z}$ and do x+{0}={x}
4. False, the subgroup $2\mathbb{Z}$ of $\mathbb{Z}$ has infinite order
5. a) <8> = {0, 8, 16}, <8> + 1= {1, 9, 17}, <8> + 2, <8> + 3, ... <8> + 7. Left and right cosets the same since addition is commutative
b) U(8) = {1, 3, 5, 7}, <3> = {3, 1}
<3>
<3> *5 = {5, 15} = {5, 7}*
Again, left and right are same since mult is commutative
c) 3Z, 3Z + 1, 3Z + 2. Left and right are same since addition is commutative
d) A_4 is all even permutations and S_4 is all permutations.
So |S_4| = 24 and |A_4|=24/2=12

SO there must be 2 cosets, the trivial one is itself.
The other coset must be odd permutations. of 4 elements. This will be true for both left and right coset
The cosets can written as A_4 *and* $(1,2)A_4$ *for left and* A_4 *and* $A_4(1,2)$ *for right*
e) Same logic as before.
For left: A_n *and* $(1,2)A_n$*, even and odd permutations respectively*
For right: A_n *and* $A_n(1,2)$*, even and odd permutations respectively*
f) $D_4 =$ *{(1),(1234),(13)(24),(1432),(13),(24),(12)(34),(14)(23)}*
$|D_4| = 8$ *so index =* $\frac{|S_4|}{|D_4|} = \frac{24}{8} = 3$ *so there are 3 cosets. We have one already*
Left:
$(12)D_4 =$ *{(12),(234),(1324),(143),(132),(124),(34),(1423)}*
$(14)D_4 =$ *{(14),(123),(1342),(243),(1243),(23),(134),(142)}*
Right:
$D_4(12) =$ *{(12),(134),(1423),(243),(34),(1324),(132),(124)}*
$D_4(14) =$ *{(14),(123),(1342),(132),(1243),(23),(143),(142)}*
g) $\mathbb{T}$ *is the set of all complex numbers such that their magnitude is 1*
The cosets of $\mathbb{T}$ *in \$\mathbb{C}^{}\$* would be $\{x \in \mathbb{C}^* : |x| = r\}$ where $r \in \mathbb{R}^+$ so there are uncountably infinite cosets. Also left and right cosets will be the same since abelian
h) There are 8 cosets total (since |H|=3)
Left:
H= {(1),(123),(132)}
(12)H={(12),(13),(23)}
(14)H={(14),(1234),(1324)}
(24)H={(24),(1423),(1342)}
(34)H={(34),(1243),(1432)}
(12)(34)H={(12)(34),(143),(243)}
(13)(24)H={(13)(24),(142),(234)}
(14)(23)H={(14)(23),(124),(134)}
Right:
H={(1),(123),(132)}
H(12)={(12),(13),(23)}
H(14)={(14),(1423),(1432)}
H(24)={(24),(1243),(1324)}
H(34)={(34),(1234),(1342)}
H(12)(34)={(12)(34),(134),(234)}
H(13)(24)={(13)(24),(124),(243)}
H(14)(23)={(14)(23),(142),(143)}

6. The left cosets are all invertible matricies that have that specific determinant. For example let $H = SL_2(\mathbb{R})$ and let $X \in GL_2(\mathbb{R})$ with det(X)=x. Then the left coset XH= {N : det(N)=det(X)=x}.

The index is therefore infinite (uncountably infinite since the determinant can be any number in $\mathbb{R}^*$)

7. a=4 and n=15, they are coprime. $\phi(15) = 4$ so $4^4 = 216$ and 216 mod 15 = 1 which verifies the statement $4^4 \equiv 1 \mod 15$.
8. Theorem: $a^{p-1} \equiv 1 \text{mod } p$ if p prime.
   Let the given be true. Let p=4n+3 be prime and $\exists$ a solution to x^2 $\equiv$ -1 mod p
   Then $a^{p-1} = a^{4n+2} = a^2a^{4n} \equiv -1(a^{4n}) = -1(a^2)^{2n} \equiv -1(-1)^{2n} = -1$
   But then this implies that $a^{p-1} \equiv -1 \text{mod } p$ not **1 mod p** as the theorem states so this is a contradiction therefore the equation must not have a solution